Strengthening Participatory Organisation (SPO)

Regional Office, Hyderabad, Sindh, Pakistan

Date: September 29, 2026

To Whom It May Concern / EPICS in IEEE Committee,

Subject: NPO Support Letter for Lokesh Kumar's Research Proposal: "Bridging the Digital Literacy Gap in Rural Agri-Logistics"

As the Program Manager for Sindh at the Strengthening Participatory Organisation (SPO), I am writing this letter in enthusiastic support of the research proposal submitted by Mr. Lokesh Kumar (Student ID: 2k22-SE-42, Software Engineering Department, Sindh Agriculture University, Tandojam) under the academic supervision of Prof. Dr. Bhawani Shankar Chowdhry.

1. Involvement in Design and Deployment

SPO has closely collaborated with rural agricultural communities across Lower Sindh (including Tharparkar, Badin, and Mirpurkhas). We have actively partnered with Mr. Lokesh Kumar during the field-testing phase of this initiative—specifically across crucial transit corridors like the Mithi-Hyderabad route. Our team assisted in coordinating with local farming communities, agricultural cooperatives, and commercial truck operators to validate the real-world operational challenges of post-harvest transit losses, which exceed 30% for perishable crops. SPO provided ground insights into local socio-economic hurdles, ensuring that the system's trilingual NLP engine (supporting Dhatki, Sindhi, and Urdu) and sub-second voice advisory features are directly tailored to the needs of marginalized, low-literacy rural transport operators.

2. Alignment with SPO's Mission

At SPO, our core mission is to foster self-reliant, resilient, and empowered rural communities through sustainable development, poverty alleviation, and equitable resource management. Mr. Lokesh Kumar's project—*Bridging the Digital Literacy Gap in Rural Agri-Logistics*—directly advances our mission by:

Mitigating Economic Losses: Addressing unbudgeted diesel deficits and structural transport inefficiencies that disproportionately burden smallholder farmers and rural supply chains in Sindh.

Empowering Vulnerable Workers: Overcoming digital and language barriers for rural drivers who cannot navigate standard English or Urdu interfaces.

Promoting Sustainable Infrastructure: Enhancing regional food security by minimizing perishable crop spoilage between rural agricultural belts and major urban markets like Hyderabad and Karachi

3. Plan for Solution Sustainability After Deployment

To ensure long-term impact and sustainability following the deployment of the live platform (https://agri-idas.tech), SPO has structured a comprehensive continuity plan:

Community Ownership & Training: SPO will facilitate grassroots workshops through our regional network in Sindh to train local transport unions, driver associations, and agricultural middlemen on utilizing the audio-navigational tool.

Institutional Integration: We will bridge connections between the research team and local agricultural market committees (Sabzi Mandis) to maintain open-source routing maps (OSRM and OpenStreetMap centerlines).

Ongoing Field Monitoring: SPO field officers will continue to monitor telemetry-driven safety and logistical metrics—such as surface friction and temperature matrices—ensuring the technology remains adaptive, maintained, and deeply integrated within the regional economy.

We strongly endorse Mr. Lokesh Kumar's project for EPICS in IEEE funding and partnership support, as it offers a transformative technological solution to deep-rooted rural challenges in Sindh.

Sincerely,

Shewa Ram Suthar

Program Manager, Sindh

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